

Gas Law Bench

Four named gas laws, one equation, and the one scale on which a temperature ratio is allowed to mean anything.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/gas-law-bench/

1 Before you open anything

Answer from what you already know, with the page closed. The point is to have a committed number on paper that the demonstration can later confirm or overturn.

PREDICTION

A sealed sample of gas occupies 24.0 L at 20.0 °C. You warm it to 40.0 °C, holding the pressure and the amount of gas constant. What volume does it now occupy?

Write the reasoning you used in one sentence — including the arithmetic you did with the two temperatures.

2 Setting three variables and letting the fourth follow

Open the demonstration. It starts on the **Charles: V–T** preset, so volume is on the vertical axis. Whichever variable sits on the vertical axis is the *dependent* one: it is solved from the other three, and its slider is locked.

1. Click **Set to STP**. Read the five boxes in the **This state** panel and fill in row 1.
2. Drag the **Pressure P** slider to **2.00 atm** (arrow keys nudge it 0.01 at a time). Fill in row 2.
3. Drag the same slider to its far right end, **4.00 atm**. Fill in row 3.
4. Click **Set to STP** again, then drag the **Amount n** slider to **2.00 mol**. Fill in row 4.

Setting	Volume (L)	Temp (K)	Temp (°C)	Molar volume V/n	PV / (nT)
STP					
P = 2.00 atm					
P = 4.00 atm					
n = 2.00 mol					

a. Which slider was locked the whole time, and what tag appeared next to its label? Say in one sentence why that variable could not be set independently.

b. One column is identical in all four rows. Name it, give its value and units, and say what that number is called.

c. Doubling the pressure from 1.00 to 2.00 atm did one thing to the volume; doubling the amount from 1.00 to 2.00 mol did another. State both effects using the numbers from your table.

The **Set to STP** button uses 0 °C and 1 atm, the convention still used in most high-school courses, which is why the molar volume comes out near 22.41 L/mol. IUPAC has defined STP as 0 °C and exactly 1 bar since 1982, where the ideal molar volume is 22.71 L/mol instead.

3 The relationship between volume and temperature

Stay on the **Charles: V–T** preset and click **Set to STP** first, so P = 1.000 atm and n = 1.000 mol are the two held variables. The Temperature slider is always in kelvin; you do not have to land on the target exactly, so record whatever the **Temp (K)** box actually reads.

1. Drag the **Temperature T** slider until the **Temp** box in kelvin reads close to 150 K. Record the reading, the **Temp** box in °C, and the **Volume** box.
2. Repeat near 250 K, 350 K, 450 K and 550 K.
3. Fill the last two columns with your own division. Carry four significant figures.

Target	Temp (K) as read	Temp (°C)	Volume (L)	$V \div T(\text{K})$	$V \div T(^{\circ}\text{C})$
150 K					
250 K					
350 K					
450 K					
550 K					

a. One of your two computed columns is constant and one is not. State the rule relating V and T in your own words, and say which temperature scale it requires.

b. The **Temperature axis** control opens on **Celsius (°C)**. Looking at the plot, read off the temperature at which the line reaches $V = 0$, and copy the label the demonstration prints on the dashed red line.

c. Now click **Kelvin (K)**. The gas has not changed. Describe what changed about the line's position relative to the origin, and explain why that makes the kelvin scale the one that ratios are allowed on.

d. Return to your Part 1 prediction. Recalculate it using your rule from (a). If your prediction was different, identify the specific assumption that failed.

4 Two states, and where the Celsius shortcut breaks

The **Two points on the same curve** panel at the bottom of the page holds two real states of the same gas: state 1 is the filled dot, state 2 the open dot.

1. With the **Charles: V–T** preset selected, click **Set to STP**. State 1 is now at 0.00 °C.
2. Check that the **Second state: T₂** readout is near 373 K (about 100 °C), and drag it there if it is not. Copy both rows of the comparison table.

State	P (atm)	V (L)	n (mol)	T (K)	T (°C)	PV/nT
State 1						
State 2						

- a. Below the table, record the values given for **V / T, state 1** and **V / T, state 2**, with units. Then say what the row labelled **Held constant between the two states** reports.

- b. Read the panel that begins **Now redo it in Celsius**. Copy exactly what it gives for **V / T(°C)** at state 1, then explain in one sentence why that is the answer and not a number.

Now work the equation backwards, the way a question would ask you to.

1. Click **Set to STP** once more.
2. In **Vertical axis**, choose **T — temperature**. Temperature is now the dependent variable and its slider locks; volume becomes settable.
3. Drag the **Volume V** slider to **30.0 L**.

- c. Record the **Temp** readings in K and in °C. Then, on paper, show the rearrangement of $PV = nRT$ that produces that temperature from $P = 1.000 \text{ atm}$, $V = 30.0 \text{ L}$ and $n = 1.000 \text{ mol}$.

d. A common shortcut is to reach for a named law by matching the two quantities a problem mentions. Which of the four presets would you have reached for in step (c), and why does going back to $PV = nRT$ work when no named law fits the wording?

5 Solve these without the demonstration

Close the page or look away from it. Use $R = 0.08206 \text{ L}\cdot\text{atm}/(\text{mol}\cdot\text{K})$ and show your working. Afterwards, reopen the demonstration and check yourself; where you disagree, find the step that went wrong rather than just correcting the answer.

1. Find the volume occupied by 1.00 mol of an ideal gas at 0.500 atm and 350.0 K. Give your answer to four significant figures.

2. A sample of 0.500 mol of gas is held at a constant 2.00 atm. Find its volume at 350.0 K, then find its volume after it is cooled to 250.0 K. Do the second part twice: once from $PV = nRT$ directly, and once as a ratio from your first answer. The two methods must agree.

3. Someone argues: "A gas at 20.0 °C is warmed to 40.0 °C at constant pressure and amount, so the temperature doubled and the volume doubles." Explain what is wrong with the argument, and give the factor by which the volume actually changes.

4. A rigid sealed steel cylinder reads 3.00 atm at 25.0 °C. It is left in a vehicle and warms to 60.0 °C. Find the new pressure. Then state which variable being fixed by the word “rigid” makes this a two-variable problem at all.

6 On your own

Nothing here is graded, and the demonstration does considerably more than this worksheet used.

- Select **Boyle: P–V** and read the note the two-state panel prints for that view. It is the only preset with no temperature in it anywhere. Work out what that means for how long a Celsius habit can go undetected.
 - Select **Avogadro: V–n** and watch the **Molar volume V/n** readout as you move the Amount slider. Decide what the slope of that line is, physically, and check it against the readout.
 - Shrink the volume until the note at the bottom of **This state** changes colour and wording. Record the molar volume at which it first warns you, and read what it says about CO₂ and NH₃ against He and N₂.
 - Use **Pause particles** and the cartoon cylinder: raise T at fixed V, then shrink V at fixed T, and describe the two different reasons the walls get hit more often.
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